Linear Algebra (MT-1004)

Homework#12

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Question 1

Let $T : \mathbb{P}_2 \to \mathbb{R}^3$ be a linear transformation defined by

$$T\left(a + bx + cx^2\right) = \begin{bmatrix} a+b \\ b+c \\ a+c \end{bmatrix}.$$

Find the matrix representation of T with respect to the standard basis $\{1, x, x^2\}$ of $\mathbb{P}_2$.